Babar Khan Durrani

Robotics & Intelligent Systems Undergraduate
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Professional Summary

Robotics and Intelligent Systems undergraduate specializing in Artificial Intelligence, Machine Learning, Computer Vision, and Generative AI. Experienced in developing end-to-end AI systems spanning computer vision, deep learning, RAG, NLP, model serving, and production-oriented inference. Hands-on experience with PyTorch, Scikit-learn, OpenCV, YOLO, FastAPI, PostgreSQL, Docker, ONNX Runtime, and locally hosted LLMs. Strong interest in building practical AI systems that combine intelligent perception with reliable software engineering.

Technical Skills

Programming: Python, C++
AI & Machine Learning: PyTorch, Scikit-learn, NumPy, Pandas, Matplotlib, Seaborn
Computer Vision: OpenCV, YOLO, Ultralytics, YOLOv8-Seg, ByteTrack
Generative AI & NLP: RAG, Sentence Transformers, Ollama, Local LLMs, ChromaDB, Embeddings, Semantic Search, n8n
AI Engineering & Deployment: FastAPI, REST APIs, SQL, Docker, Docker Compose, ONNX Runtime
Robotics & Hardware: ESP32, Arduino, Proteus, PCB Design
Networking & Simulation: Cisco Packet Tracer, MATLAB/Simulink
Business: B2B & B2C Sales, Client Negotiation

Education

Bahria University, Islamabad
BS Robotics and Intelligent Systems — Sep 2023 – Sep 2027
CGPA: 3.11/4.00
Relevant Coursework: Artificial Intelligence, Machine Learning, Computer Vision, Embedded Systems, Robotics, Automation, Control Systems

Work Experience

Bridgeway Telecom Pvt. Ltd., Rawalpindi — Jan 2023 – Sep 2023
Customer Service Representative

- Managed B2B customer interactions in the UK sector, assisting businesses with electricity and gas contracts.
- Trained and mentored new team members on customer handling, communication, and sales procedures.

EDM Solutions Pvt. Ltd., Rawalpindi — Sep 2023 – Nov 2023
Appointment Setter

- Conducted B2B outreach to construction and remodeling companies to promote Facebook advertising services.
- Scheduled qualified appointments for senior sales executives and maintained professional communication with prospects.

Internship Experience

Embotics (Pvt.) Ltd., Islamabad July 2026 – Sep 2026
AI & ML Intern

- Contributed to an AI Teacher Robot providing interactive educational assistance using curriculum-based textbooks from Grades 2–9.
- Processed textbooks by extracting, cleaning, chunking, and organizing content into a vector database for semantic retrieval.
- Developed a Retrieval-Augmented Generation (RAG) pipeline using locally hosted LLMs through Ollama to generate curriculum-grounded responses.
- Deployed and tested the AI system on Raspberry Pi, evaluating local inference performance and hardware constraints.
- Integrated and tested audio-based interaction for spoken responses and more natural human-robot interaction.
- Tested the system across multiple grade levels and subjects to evaluate retrieval quality, response relevance, and reliability.

Projects

RoboSight — Software-Only Robotic Visual Perception System
Built a robotic perception pipeline that converts monocular RGB images and video into structured 2.5D scene representations using YOLOv8-Seg, Depth Anything V2, and ByteTrack. Implemented instance-level depth alignment, object tracking, spatial relations, motion estimation, scene-graph generation, and a rules-based query layer for robot-style scene understanding.

VisionServe — Production Computer Vision Inference Platform
Developed a production-style computer vision inference service using FastAPI, PostgreSQL, Docker, PyTorch, and ONNX Runtime. Implemented validated image inference, model versioning, structured logging, database persistence, API testing, Docker Compose deployment, and native PyTorch versus ONNX Runtime benchmarking for CPU inference.

Local AI Research Assistant
Developed a fully local RAG-based research assistant for semantic search and question answering over research papers. Implemented document chunking, BGE embeddings, ChromaDB vector retrieval, relevance filtering, source tracking, automated paper summarization, and two-paper comparison using a locally hosted LLM.

Visual Anomaly Inspection using Deep Feature Representations
Developed an unsupervised industrial anomaly detection system using ImageNet-pretrained ResNet-50 features and nearest-neighbor scoring. Evaluated global, patch-level, and ROI-aware approaches on MVTec AD 2, achieving up to 0.847 ROC-AUC and 0.969 Average Precision on Sheet Metal using PyTorch, OpenCV, and Scikit-learn.

Resume Analyzer — ATS Score & Role-Fit Feedback Tool
Built an offline NLP tool using sentence embeddings (Sentence-BERT) to semantically match resumes against 30+ job roles and generate ATS compatibility scores. Implemented PDF parsing, automated section detection, experience-level classification, and adaptive skill-gap suggestions with per-skill relevance scoring.

Aquabot — WaveRunner (Autonomous/RC Boat Competition)
Designed and built a remote-controlled aquatic robot for a competitive goal-scoring event in a

pool environment. Used Arduino for control logic, brushless motors with ESCs for propulsion, an IMU sensor for stability and orientation feedback, and a 2.4GHz RC system for real-time control.

Languages

Pashto (Native)
Urdu (Fluent)
English (Fluent)
Punjabi (Intermediate)